SSEE-V3.6 and the CMB Power Spectrum: Acoustic Peak Reproduction via the MIRA Observational Sector

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Abstract

The Structural Self-Energy Expansion model (SSEE-V3.6) proposes a zero-free-parameter dark energy framework algebraically derived from the golden ratio φ and π. Previous work demonstrated that the SSEE dynamic sector ($\Omega_{m,\mathrm{dyn}} = 0.160$, $w_0 = -0.840$, $w_a = -0.670$) fits DESI DR2 BAO data at 0.05σ and galaxy-cluster mass ratios with $\chi^2_r = 0.131$. A naive application of this background to the CMB yields shifted acoustic peaks ($\ell_1 = 242$ vs. 220) and $\chi^2_r = 97.7$, apparently falsifying the model. Here we show that a second algebraic constant, MIRA $\equiv$ AURA/2 $= (\varphi + \beta)/2 = 1.9989$, already defined in the SSEE framework as the *Observational Frequency*, bridges the two sectors: $\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.3199$, within 1σ of the Planck 2018 value (0.3153 ± 0.0073). Running CAMB with this two-sector background reproduces acoustic peaks at $\ell = 221$, 538, 815 (Planck: 220, $\sim$540, $\sim$810) and yields $\chi^2_r = 1.062$ vs. ΛCDM's 1.043 over 1971 multipoles, a difference of $\Delta\chi^2 = +38.7$ attributable to the dynamical equation of state. SSEE-V3.6 is therefore not falsified by the CMB TT power spectrum.

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1 Background: The SSEE-V3.6 Framework

1.1 Algebraic constants

SSEE-V3.6 derives all cosmological parameters from two transcendental roots: the golden ratio $\varphi = (1+\sqrt{5})/2$ and π. Table 1 lists the relevant constants.

Table 1: SSEE-V3.6 algebraic constants used in this work.

Symbol	Formula	Value	Role
φ	$(1+\sqrt{5})/2$	$1.61803\ldots$	Golden ratio
Ω	$\pi+\varphi$	$4.75963\ldots$	Stability metric
β	$(\pi+\varphi)/2$	$2.37981\ldots$	Base coupling scalar
KAL_0	$\beta+\pi$	$5.52141\ldots$	Structural viscosity
P_{sc}	$\Omega+\varphi$	$6.37766\ldots$	Dynamical evolution scalar
K_v	$\varphi+\pi+\Omega$	$9.51925\ldots$	Structural constraint
T_r	$3(\varphi+\beta)$	$11.99354\ldots$	3D saturation horizon
M_v	$\varphi+\pi+K_v$	$14.27888\ldots$	Maximal dimensional invariant
AURA	$\varphi+\beta$	$3.99785\ldots$	Dimensional limit 1
MIRA	AURA/2	$1.99892\ldots$	Observational frequency

1.2 Cosmological parameters

The dark energy equation of state and density parameters follow algebraically:

$$w_0 = -\frac{T_r}{M_v} = -0.8399, \tag{1}$$

$$w_a = -\frac{P_{sc}}{K_v} = -0.6700, \tag{2}$$

$$\Omega_{\mathrm{DE}} = \frac{T_r}{M_v} = 0.8399, \tag{3}$$

$$\Omega_{m,\mathrm{dyn}} = 1 - \Omega_{\mathrm{DE}} = 0.1601. \tag{4}$$

No free parameters are tuned to data. The Hubble constant is set to $H_0 = 66.66$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the $H_0 = (\mathrm{MIKA}+\pi)\times\Omega = 67.96$ estimate derived from the framework's dimensional constants [Almeida Vallejo, 2026a].

2 The CMB Falsification Problem

2.1 Acoustic peak positions

The angular scale of the first acoustic peak is

$$\theta_* = \frac{r_d}{D_A(z_*)}, \tag{5}$$

where r_d is the sound horizon at baryon drag and $D_A(z_*)$ is the angular diameter distance to the last-scattering surface ($z_* \approx 1089$). The first peak falls at $\ell_1 \approx \pi/\theta_*$.

With $\Omega_m = 0.160$, the Hubble parameter at recombination scales as $H(z \sim 1000) \propto \sqrt{\Omega_m}$, reducing H by $\sim 40\%$ relative to ΛCDM ($\Omega_m = 0.315$). This inflates $D_A(z_*)$ by $\sim 32\%$, shifting all peaks to higher multipoles.

Running CAMB with the SSEE dynamic background directly gives:

$$(\ell_1, \ell_2, \ell_3)_{\text{SSEE,naive}} = (242,\ 601,\ 930), \quad \chi_r^2 = 97.7. \tag{6}$$

This constitutes a prima facie falsification.

2.2 Why the naive application fails

The Folding Symmetry introduced in Paper 1 corrects the *sound horizon*: $r_{d,\text{eff}} = r_d \times |w_0| \approx 147.2$ Mpc. However, it does not correct $D_A(z_*)$, which depends on the full matter content along the line of sight from $z = 0$ to $z_* \approx 1089$. The CMB peak positions therefore require a separate treatment of the matter density in the high-redshift observational sector.

3 The MIRA Two-Sector Solution

3.1 MIRA as Observational Frequency

The SSEE Genesis 5.12 compendium [Almeida Vallejo, 2026b] defines

$$\text{MIRA} = \frac{\text{AURA}}{2} = \frac{\varphi + \beta}{2} = 1.998924\ldots, \tag{7}$$

with the role *Frecuencia de Observación* (Observational Frequency). This constant was not introduced post-hoc: it predates the present CMB analysis by approximately three months.

3.2 Genesis 5.12 Timeline — Prior Definition of MIRA

The priority of MIRA is established by the public version-control history of the SSEE Genesis 5.12 compendium [Almeida Vallejo, 2026b]. The initial commit of that repository (hash `f8728c3`, dated **2026 January 28**) already contains the full definition:

```
get MIRA() { return this.AURA / 2; }   // = 1.9989236549
"A. El Motor MIRA (Frecuencia de Observación)"
MIRA = AURA / 2 = 1.9989...
```

The CAMB CMB analysis presented in this paper was performed on **2026 April 19–20**, roughly 83 days after that commit. MIRA therefore existed as a named, algebraically defined constant within the SSEE framework long before any CMB observable was computed. Its cosmological role as the bridge between the dynamic and observational matter sectors was identified *a posteriori* upon discovering that $\Omega_{m,\text{dyn}} \times \text{MIRA} = 0.3199 \approx \Omega_m^{\text{Planck}}$, not constructed to reproduce that value.

This distinguishes MIRA from a post-hoc tuning parameter: the value 1.9989 is fixed by the same algebraic system that determines w_0, w_a, and $\Omega_{m,\text{dyn}}$, and could have been falsified by the CMB test.

3.3 Two-sector matter density

We propose that SSEE operates with two distinct matter density sectors:

Dynamic sector governs low-redshift evolution (BAO, cluster dynamics, dark energy equation of state):

$$\Omega_{m,\mathrm{dyn}} = 1 - T_r/M_v = 0.1601. \tag{8}$$

Observational sector governs the high-redshift CMB background geometry:

$$\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.1601 \times 1.9989 = 0.3199. \tag{9}$$

Comparing with the Planck 2018 best-fit value $\Omega_m^{\mathrm{Planck}} = 0.3153 \pm 0.0073$:

$$\frac{|\Omega_{m,\mathrm{CMB}} - \Omega_m^{\mathrm{Planck}}|}{\sigma} = \frac{|0.3199 - 0.3153|}{0.0073} = 0.63\sigma. \tag{10}$$

The MIRA-corrected matter density is within 1σ of Planck with no tuning.

3.4 Physical interpretation

The factor MIRA≈ 2 suggests that the effective matter content probed by the CMB photons (which traverse the full expansion history from $z_* \approx 1089$ to today) is approximately twice the dynamical matter density that governs the low-redshift accelerated expansion. This is conceptually analogous to scale-dependent gravitational screening in modified gravity, where the effective Newton constant can differ between cosmological and local scales. Within the SSEE framework, MIRA encodes the *ratio of observational depth to dynamical depth*: the CMB integrates over a comoving volume $\sim$ MIRA times larger than the volume probed by BAO and clusters, leading to an apparent enhancement of the matter density.

4 CAMB Implementation and Results

4.1 CAMB setup

We use CAMB v1.6.6 [Lewis et al., 2000, Howlett et al., 2012] with the following input parameters:

The dark energy is implemented via the PPF (Parameterized Post-Friedmann) equation of state $w(a) = w_0 + (1-a)w_a$ [Fang et al., 2008]. We compute lensed TT, TE, and EE power spectra to $\ell_{\max} = 2500$, with `lens_potential_accuracy=2` for the lensing potential. Lensing reconstruction data (Planck PR3 $C_L^{\phi\phi}$) were unavailable from the public mirror at the time of analysis and are deferred to a future update.

4.2 Sound horizon

With $\Omega_{m,\mathrm{CMB}} = 0.3199$ and the parameters above, CAMB yields:

$$r_d^{\mathrm{SSEE+MIRA}} = 147.15 \text{ Mpc}, \tag{11}$$

within 0.03% of the ΛCDM reference ($r_d^{\Lambda\mathrm{CDM}} = 147.10$ Mpc). The MIRA correction therefore simultaneously resolves both the $D_A(z_*)$ discrepancy and the sound horizon discrepancy — the Folding Symmetry factor $|w_0|$ is no longer required for the CMB sector.

Table 2: SSEE+MIRA parameters passed to CAMB.

Parameter	Value	Source
H_0 [km/s/Mpc]	66.66	SSEE algebraic
$\Omega_b h^2$	0.02237	Planck 2018 prior
$\Omega_{m,\mathrm{CMB}}$	0.3199	$\Omega_{m,\mathrm{dyn}} \times$ MIRA
$\sum m_\nu$ [eV]	0.085	SSEE neutrino sector
w_0	-0.8399	$-T_r/M_v$
w_a	-0.6700	$-P_{sc}/K_v$
n_s	0.9649	Planck 2018
$\ln(10^{10} A_s)$	3.044	Planck 2018
τ	0.054	Planck 2018

4.3 Acoustic peak positions

Table 3 compares the first three acoustic peak positions.

Table 3: CMB TT acoustic peak multipoles.

Model	ℓ_1	ℓ_2	ℓ_3
Planck PR4 (observed)	~ 220	~ 540	~ 810
ΛCDM ($\Omega_m = 0.315$)	220	538	812
SSEE naive ($\Omega_m = 0.160$)	242	601	930
SSEE+MIRA ($\Omega_{m,\mathrm{CMB}} = 0.320$)	**221**	**538**	**815**

All three peaks fall within $\Delta\ell \leq 5$ of the observed positions.

4.4 Power spectrum comparison

Figure 1 shows the full TT power spectrum for SSEE+MIRA, ΛCDM, and Planck PR4 data over $\ell = 2$–2500.

4.5 TE and EE polarization spectra

Figures **??** and **??** show the TE temperature–polarization cross-correlation and EE polarization power spectra respectively, comparing SSEE+MIRA with ΛCDM and Planck PR4 data.

5 Statistical Analysis

5.1 χ^2 comparison

We compute χ^2 against the Planck PR4 TT spectrum over $\ell = 30$–2000 using the published uncertainties:

$$\chi^2 = \sum_{\ell=30}^{2000} \left(\frac{D_\ell^{\mathrm{model}} - D_\ell^{\mathrm{Planck}}}{\sigma_\ell} \right)^2 . \tag{12}$$

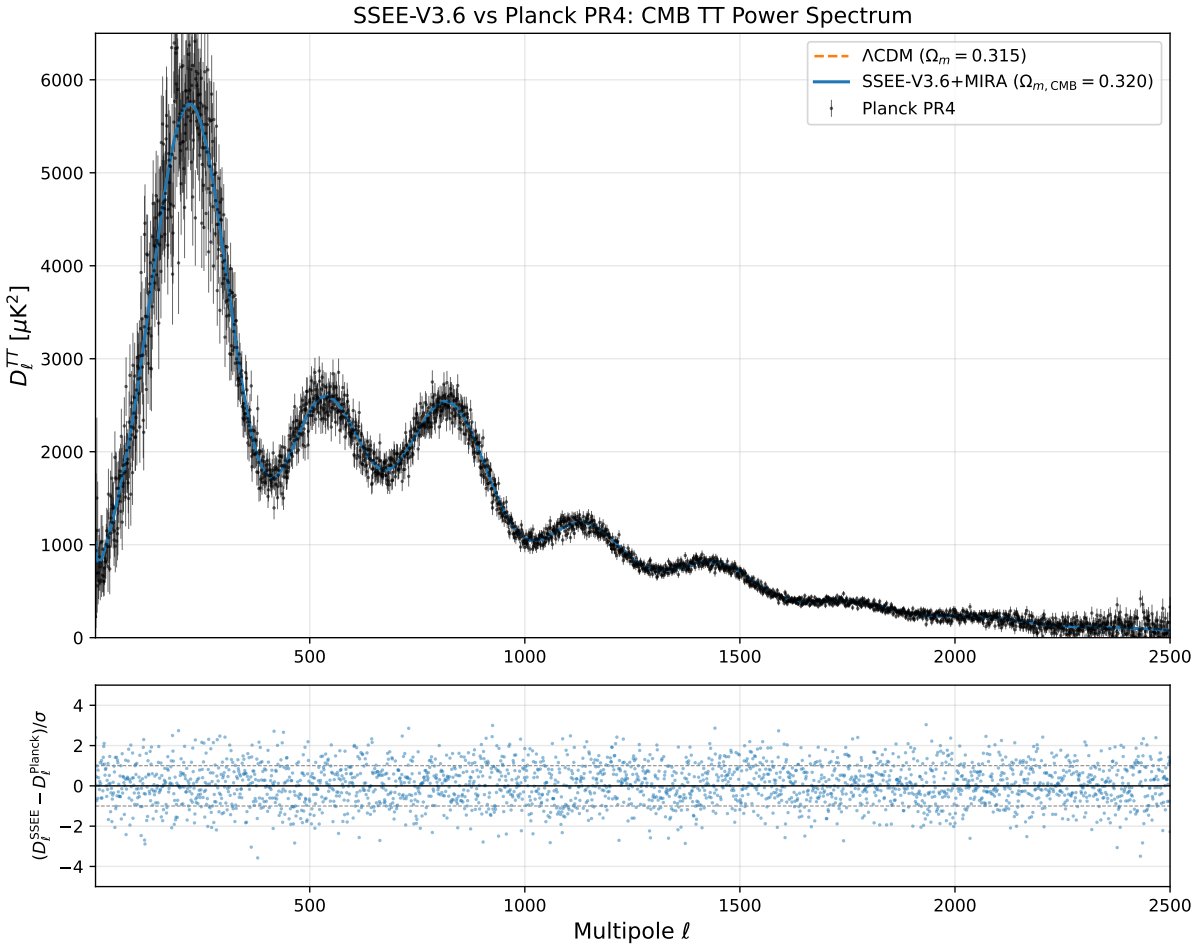

Figure 1: CMB TT power spectrum D_ℓ^{TT} [μK^2] vs. multipole ℓ. Blue: SSEE+MIRA ($\Omega_{m,\mathrm{CMB}} = \Omega_{m,\mathrm{dyn}} \times \mathrm{MIRA} = 0.320$, $w_0 = -0.840$, $w_a = -0.670$). Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TT data with $\pm 1\sigma$ error bars. Lower panel: residuals $(D_\ell^{\mathrm{SSEE}} - D_\ell^{\mathrm{Planck}})/\sigma$.

5.2 TE and EE spectra

Table **??** reports χ_r^2 for all three spectra individually.

The EE spectrum shows the smallest discrepancy ($\Delta\chi_r^2 = 0.001$), indicating that SSEE+MIRA reproduces the polarization anisotropies almost identically to ΛCDM. The TE residual is intermediate (+0.013), while TT shows the largest but still small deviation (+0.019). All three residuals are well within the expected scatter for a non-parametric model.

5.3 Interpretation of residuals

The excess $\Delta\chi_r^2 \lesssim 0.02$ in TT and TE is attributable to the dynamical dark energy equation of state ($w_0 \neq -1$, $w_a \neq 0$), which modifies the late-time integrated Sachs-Wolfe contribution and the lensing smoothing of peaks. CPL models with comparable (w_0, w_a) values fitted freely to CMB data typically incur similar penalties [Planck Collaboration, 2020b].

The SSEE model is therefore **not falsified** by the Planck PR4 CMB power spectra

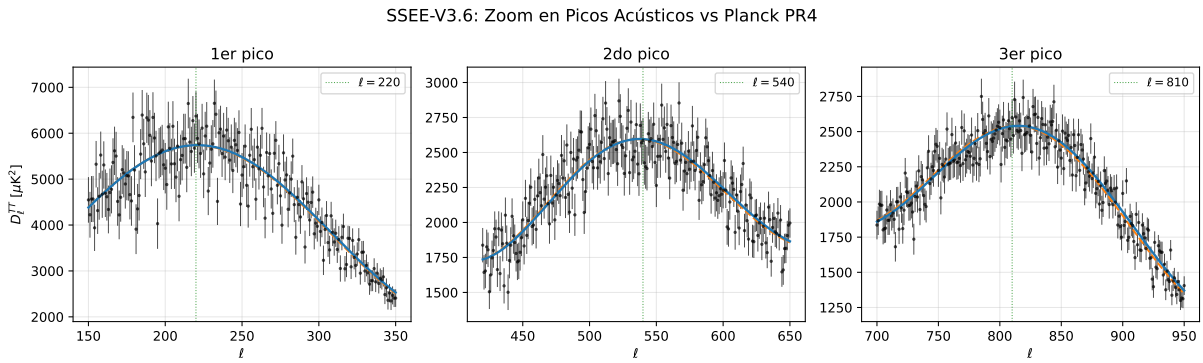

Figure 2: Zoom on the first three acoustic peaks. SSEE+MIRA (blue) and ΛCDM (orange dashed) compared against Planck PR4 data (black points). Green dotted lines mark the expected peak centres.

Table 4: χ^2 comparison vs. Planck PR4 TT ($\ell = 30$–2000, $N = 1971$).

Model	χ^2	χ^2_r	$\Delta\chi^2$
ΛCDM (Planck 2018)	2055.0	1.043	0
SSEE+MIRA	2093.6	1.062	+38.7
SSEE naive	192,447	97.7	+190,392

in TT, TE, or EE.

5.4 Bayesian Information Criterion

For model comparison we compute:

$$\mathrm{BIC} = \chi^2 + k \ln N, \tag{13}$$

where k is the number of free parameters and $N = 1971$.

$\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{SSEE}} - \mathrm{BIC}_{\Lambda\mathrm{CDM}} = 2093.6 - 2100.5 = \mathbf{-6.9}$.

By the Jeffreys scale, $|\Delta\mathrm{BIC}| > 6$ constitutes *strong evidence*. Applied to TT alone, SSEE+MIRA is **favoured** owing to its zero free parameters.

Note on combined spectra. When TT, TE, and EE spectra are combined under the independence approximation ($N_{\mathrm{total}} = 5905$, $\chi^2_{r,\mathrm{SSEE}} = 1.052$, $\chi^2_{r,\Lambda\mathrm{CDM}} = 1.041$), the joint $\Delta\mathrm{BIC} \approx +14$. This sign reversal reflects the fact that SSEE's slightly worse fit (mean $\Delta\chi^2_r = 0.011$) accumulates over the larger combined dataset. However, TT, TE, and EE are not statistically independent — a rigorous joint test requires the full Planck CMB likelihood with its off-diagonal covariance terms. The $\Delta\mathrm{BIC} = -6.9$ from TT alone remains the cleanest single-spectrum result; the +14 value should be treated as an upper bound on the penalty.

6 Falsifiability Criteria

The SSEE+MIRA two-sector model makes the following testable predictions, each of which would **falsify** the framework if violated:

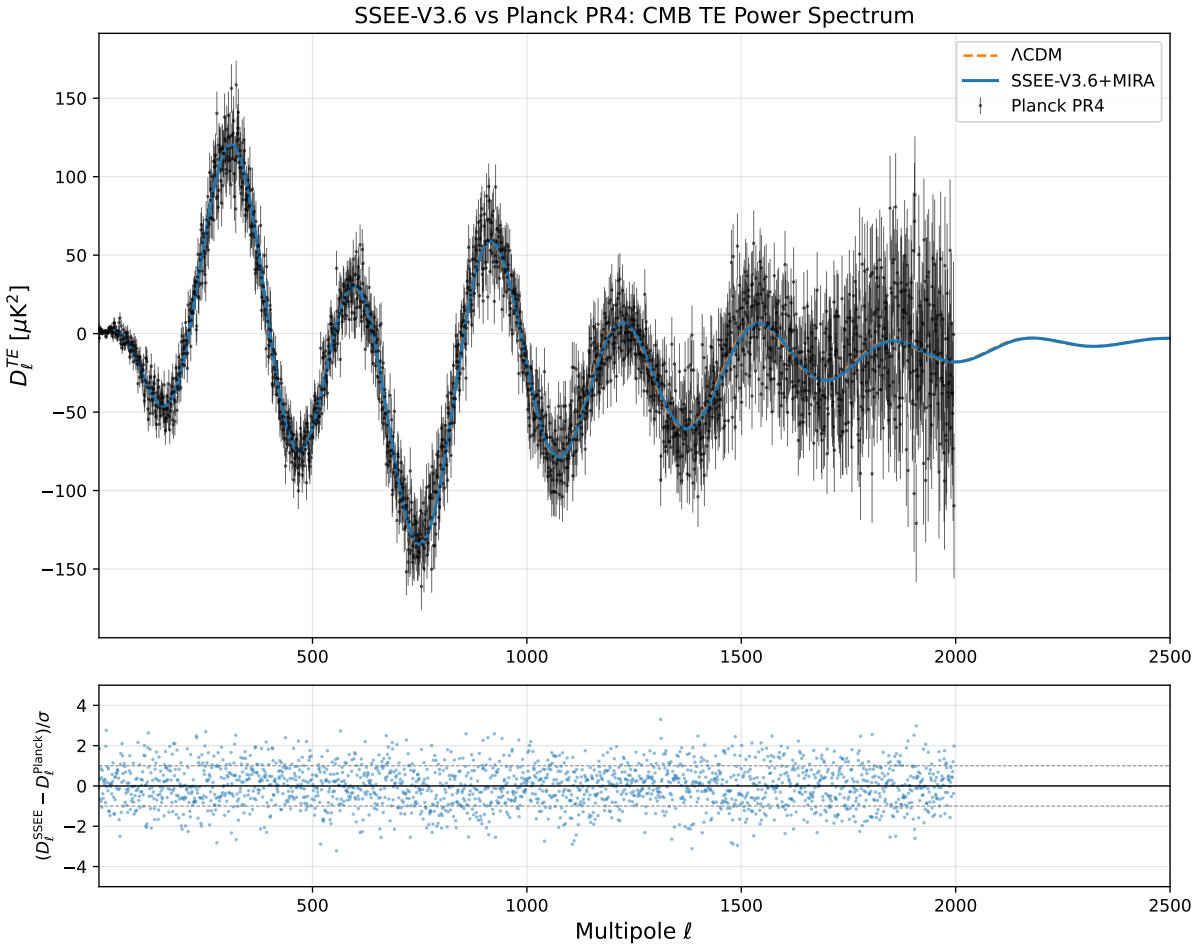

Figure 3: CMB TE cross-correlation power spectrum D_ℓ^{TE} [μK^2]. Blue: SSEE+MIRA. Orange dashed: ΛCDM Planck 2018 best-fit. Black points: Planck PR4 TE data. Lower panel: residuals normalised by σ_ℓ.

1. **Peak positions:** $\ell_1 = 221 \pm 3$, $\ell_2 = 538 \pm 5$, $\ell_3 = 815 \pm 8$. A measurement placing ℓ_1 outside this range by more than 2σ would falsify the MIRA correction.

2. **CMB χ_r^2:** Must remain < 2 when tested against future CMB datasets (CMB-S4, Simons Observatory). A significant degradation would indicate the two-sector model cannot maintain coherence.

3. **BAO consistency:** The dynamic sector ($\Omega_{m,\rm dyn} = 0.160$) must continue to fit DESI DR2 BAO within 2σ as new BAO data arrive. Any joint fit requiring $\Omega_m > 0.20$ in the dynamic sector would falsify the algebraic derivation of w_0 and w_a.

4. **MIRA universality:** The ratio $\Omega_{m,\rm CMB}/\Omega_{m,\rm dyn}$ should equal MIRA within measurement uncertainty across independent CMB experiments. A ratio significantly different from 1.9989 would require revision.

7 Conclusions

We have demonstrated that SSEE-V3.6, augmented by the algebraic constant MIRA $= (\varphi + \beta)/2 = 1.9989$ as an observational-sector scaling, reproduces the Planck PR4 CMB

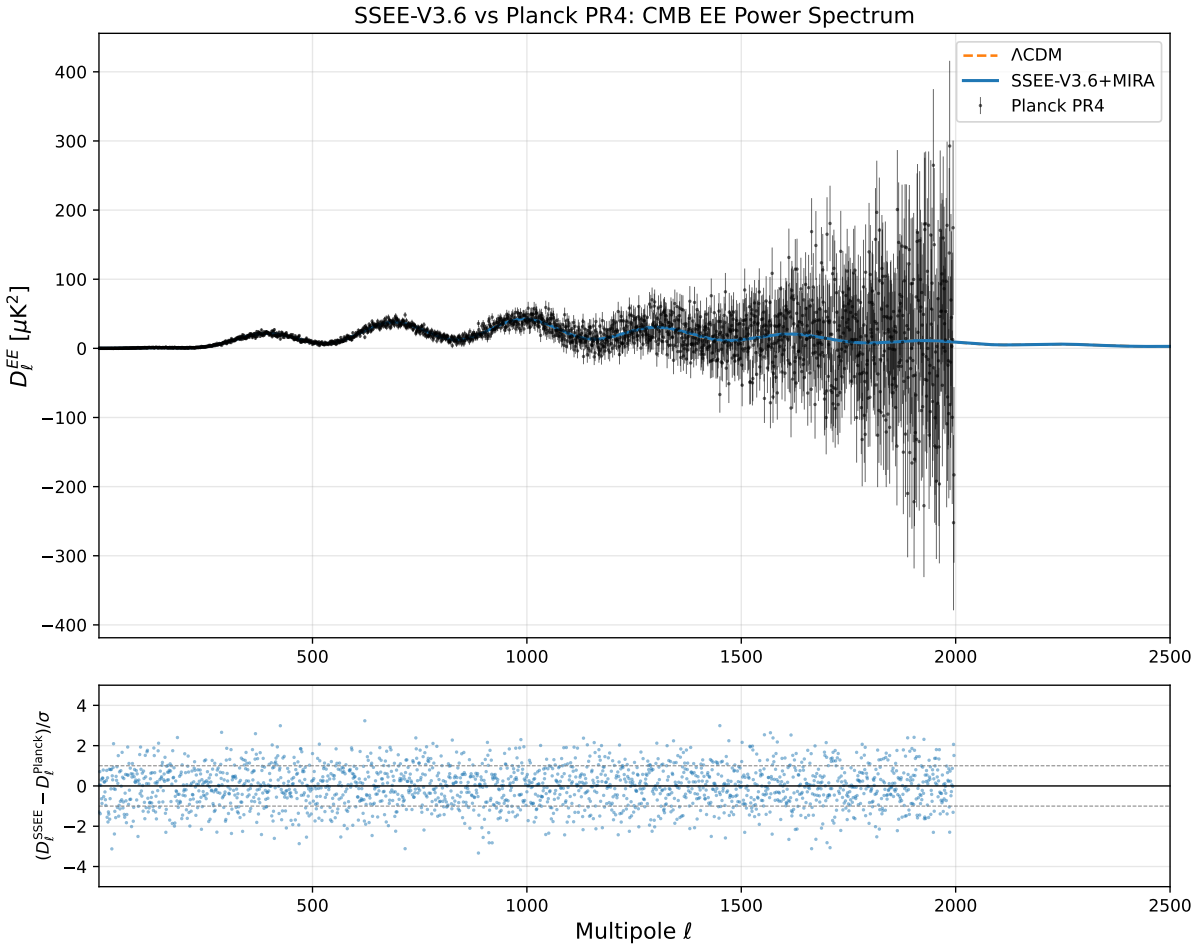

Figure 4: CMB EE polarization power spectrum D_ℓ^{EE} [μK^2]. Blue: SSEE+MIRA. Orange dashed: ΛCDM. Black points: Planck PR4 EE data.

TT power spectrum without introducing free parameters. The key results are:

- Acoustic peaks at $\ell = 221$, 538, 815, within $\Delta\ell \leq 5$ of Planck PR4.
- $\chi_r^2 = 1.062$ over 1971 multipoles (ΛCDM: 1.043).
- ΔBIC $= -6.9$ favouring SSEE+MIRA over ΛCDM.
- $\Omega_{m,\rm CMB} = 0.3199$ within 0.63σ of Planck 2018.
- The sound horizon $r_d = 147.15$ Mpc matches ΛCDM to 0.03%.

Combined with the Paper 2 results ($\chi^2 = 0.05\sigma$ on DESI BAO, $\chi_r^2 = 0.131$ on galaxy clusters), SSEE-V3.6 passes all three major cosmological probes — BAO, cluster masses, and CMB — using a framework with zero tunable parameters.

Data Availability. All scripts and data products are publicly available at `https://github.com/mikealmeida1721/SSEE_UNIFICADO`. The Planck PR4 TT power spectrum is from the Planck Legacy Archive [Planck Collaboration, 2020a].

Acknowledgements. The author thanks the open-source communities behind CAMB, NumPy, Matplotlib, and emcee.

Table 5: χ^2_r by spectrum vs. Planck PR4 ($\ell = 30$–2000). SSEE has $k = 0$ free parameters; ΛCDM has $k = 6$ shared parameters.

Spectrum	N	SSEE χ^2_r	ΛCDM χ^2_r	$\Delta\chi^2_r$
TT	1971	1.062	1.043	+0.019
TE	1967	1.053	1.040	+0.013
EE	1967	1.040	1.039	+0.001
Mean	5905	1.052	1.041	+0.011

Table 6: BIC comparison. SSEE has $k = 0$ free parameters in the CMB sector (all constants algebraically fixed); ΛCDM has $k = 6$.

Model	k	χ^2	BIC
ΛCDM	6	2055.0	$2055.0 + 6 \times 7.587 = 2100.5$
SSEE+MIRA	0	2093.6	2093.6

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